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Effects of repeated retrieval on long-term retention in a nonverbal learning task in younger children

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ABSTRACT

Many studies have shown the benefits for long term retention of repeated retrieval during learning in verbal tasks, but few have shown its effectiveness using nonverbal materials. The aim of this study was to examine whether the retention benefits of repeated retrieval extend to preschool children performing a spatial location memory task. In this task, the children first studied where eight small toys were located in a partitioned box. Then, in the repeated retrieval condition, the children were asked to put each one of the toys in its place by themselves three times successively with feedback, whereas in the repeated study condition, they were asked to put each toy in its place with the experimenter showing them the correct location. Half of the children were then immediately tested and the remaining half tested after one day. The results showed that the 5 and 6 year old children in the retrieval condition retained location memory for the toys longer than those in the study condition in a memory task involving spatial content and enactment components. These findings have deep theoretical implications for the critical role of retrieval effort in long-term retention, and highlight the efficacy of repeated retrieval for different developmental ages and tasks.

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KEYWORDS Repeated retrieval; nonverbal learning task; long-term retention; younger children

Introduction

It is important for children's adaptation to daily life to promote the long-term retention of not only to-be-remembered events and general factual information, but also where things are located in their spatial surroundings. Parents and preschool teachers often ask children where they left their toys and other items and try to teach them to put their belongings away in a specific place. It is not identity memory of particular items that is important here, but rather spatial location memory involving context, space, and action memories that are key

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to remembering locations. The enhancement of location and spatial memory can thus be extremely useful in a child's home and school environment, as well as later in life. Moreover, location memory may serve as the foundation for higher-level reasoning such as spatially rich mathematics (Spelke, Lee, & Izard, 2010), and science, technology, and engineering (Newcombe, 2010; Uttal et al., 2013). The current study focuses on whether repeated retrieval is a more effective learning method for enhancing the long-term retention of location memory in preschool children than a more passive study method.

A number of studies with adults have demonstrated that retrieval practice is one of the most effective long-term retention strategies, especially concerning verbal material (see Karpicke, Lehman, & Aue, 2014; Pastötter & Bäuml, 2014; for a review). Findings for normal adults have demonstrated that repeated retrieval promotes greater long-term retention of learning material compared to repeated studying (e.g., Roediger & Karpicke, 2006a, 2006b; see also, Hogan & Kintsch, 1971). In Roediger and Karpicke's (2006b) procedure for instance, adult participants first studied the same two passages, then in the restudy condition one of the passages was presented four consecutive times, while the remaining passage was assigned to the repeated retrieval condition and was retrieved by participants three consecutive times. Half the participants were then tested after 5 min and the others were tested after 1 week. The performance of the 5 min testing group in the restudy condition was better than in the repeated retrieval condition, whereas with the 1 week delayed testing group, performance in the repeated retrieval condition was better than that in the restudy condition. According to Rowland's (2014) review, theoretical accounts of the effectiveness of retrieval practice have emphasized the importance of desirable difficulty enhancing the relevance and memory of learned items (e.g., Bjork & Bjork, 1992). The greater effort and practice involved with having to generate answers rather than merely studying facilitates retention and recall (e.g., Carpenter & Delosh, 2006; Glover, 1989; Karpicke & Roediger, 2007). Retrieval effort may induce mediating information (e.g., Pyc & Rawson, 2010), which consequently can lead to elaborative retrieval, thereby enhancing recall (Carpenter, 2009; see also, Rowland, 2014).

As stated earlier, many studies on the testing effect have been conducted in the verbal learning area using a variety of educational contexts and materials. Similar results were also obtained for elementary school children (e.g., Goossens, Camp, Verkoeijen, Tabbers, & Zwaan, 2014; Lipowski, Pyc, Dunlosky, & Rawson, 2014; Morris, Fritz, Jackson, Nichol, & Roberts, 2005) and preschool children (e.g., Fritz, Morris, Nolan, & Singleton, 2007). Fritz et al. (2007, Experiment 2) showed an interesting finding that preschoolers as young as three and four who learned the names of toys were dramatically better at recalling these names after expanded retrieval practice than after expanded re-presentation or massed elaboration both after 1 min and after 1 day. Fritz et al. (2007) also noted that children enjoyed and kept their attention on the task in the retrieval condition

much more than they did in the repeated study condition. With the lack of other studies apart from Fritz et al. (2007), the robustness of the effects of repeated retrieval on long-term retention for preschool children remains unclear. In particular, little is known about learning in nonverbal, spatial memory, such as an object location task for preschool children.

Several studies with adults and primary school children show a powerful retrieval effect using spatial materials, such as map or route learning (e.g., Carpenter & Kelly, 2012; Carpenter & Pashler, 2007; Kelley, Carpenter, & Sjolund, 2015; Rohrer, Taylor, & Sholar, 2010). Rohrer et al. (2010) compared the benefits of restudy and retrieval in fourth and fifth graders who had to learn regions or cities on fictional map locations. After 1 day, they were tested on the learned material using both an identical task and a transfer task. In both tasks, the children performed better after retrieval practice than after restudying. According to Kelley et al. (2015), retrieval could enhance route memory compared to studying, but only when participants were given feedback about the correct answer prior to an incorrect response, whereas retrieval caused worse performance compared to studying when feedback occurred after an incorrect response (see also Maxwell, Masters, Kerr, & Weedon, 2001). When verbal materials involving identity memory were used, however, many studies show that retrieval errors do not impair learning, as long as correct answers are provided after a retrieval attempt (e.g., Butler & Roediger, 2008; Kulhavy & Stock, 1989; Pashler, Cepeda, Rohrer, & Wixted, 2005). The ideal conditions for enhancing the testing effect in a spatial location task appear to be different. In consideration of the findings that errorless retrieval in spatial memory promotes retention (Maxwell et al., 2001), we include a feedback component to prevent incorrect responses during initial study immediately before the retrieval/studying phase.

In summary, we examined the effectiveness of retrieval vs. restudying schedules on long-term retention for 5 and 6 year old children in a nonverbal location task with small toys. The experimental procedure consisted of three phases. In the first phase, children were taught where each small toy (such as a toy spoon or a banana) had been located in a partition in a box. At this time, the experimenter checked whether the children could overtly show the place of each toy by pointing to the correct area. The children were then told the correct answer to promote errorless learning. In the second phase, all the children were asked to perform the action (e.g., put the banana toy in its place) in both the repeated retrieval and restudy conditions. In the repeated retrieval condition, the experimenter asked them to put one of the toys in its place themselves with feedback, whereas in the repeated study condition, they were asked to put each toy in its place jointly, with the experimenter guiding them. In the third phase, half the children took the immediate test and the remaining half was tested after one day.

As stated before, this study had an enactment component (e.g., put the banana toy in its place) in both the retrieval and restudy conditions. Basically,

memories formed by performing an action during encoding produce better performance than memories based on information watched passively (Cohen, 1989; Nilsson, 2000; Zimmer et al., 2001). Thus, the procedure during the second phase in this study was similar to a self-performed task (for review, Nilsson, 2000). More importantly, although all the children performed the action both in the retrieval and studying conditions, we focused on the difference between self-testing and experimenter-induced studying on retention. The self-test manipulation in the repeated retrieval condition would require more retrieval effort (e.g., Bjork & Bjork, 1992) and more strongly associate the objectives with the context (such as perceptual properties) compared to the repeated study condition, which would support the mediated aspect of repeated retrieval (e.g., Pyc & Rawson, 2010). Based on Roediger and Karpicke's (2006b) findings of the efficacy of repeated retrieval for long-term retention, but not immediate recall, we predicted that no differences on the immediate test would be observed for the repeated retrieval and repeated study conditions, whereas on the delayed test the performance in the repeated retrieval condition would be better than that in the repeated study condition. If repeated retrieval produces the better performance after one day, this would be the first demonstration that we know of that the testing effect can be found with a spatial task in preschool children.

Methods

Participants

Sixty children from 5 to 6 years old (22 girls and 38 boys; M = 71.32 months (SD = 3.05), age range = 61–80 months)) participated individually. Of the 60 children, 30 children were randomly assigned to the immediate test group and the remaining 30 were assigned to the delayed testing group. Children in the immediate and delayed test groups were from 5 years old (immediate; 15 children, M = 66.40 (SD = 2.85), delayed; 15 children, M = 65.73 (SD = 3.52)) to 6 years old (immediate; 15 children, M = 76.27 (SD = 3.01), delayed; 15 children, M = 76.87 (SD = 2.83)).

Design

The variables were Learning condition (Retrieval/Studying) and Retention interval group (Immediate/Delayed). Learning condition was the within-participants factor and Retention interval was the between-participants factors. The dependent variables were the percent correctly recalled.

Materials

(1) Study: all children were given eight small toy items (that fit in a 12 by 10 cm partition) and shown their corresponding place out of 10 possible locations. Two

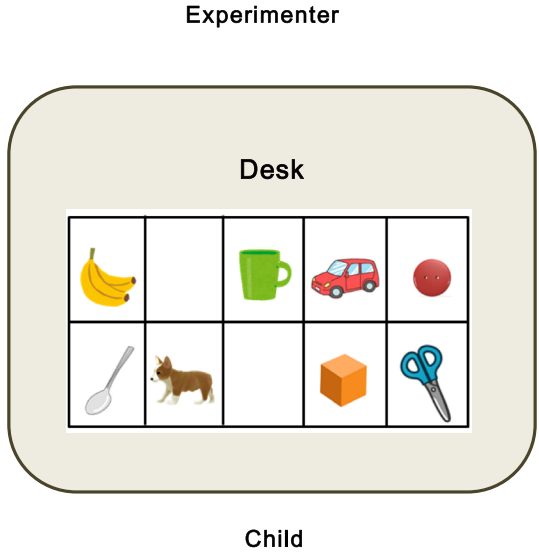


Figure 1. Materials used in the experiment.

of the ten possible locations had nothing in them (see Figure 1). All the children were presented with the same 24-by-50 cm box on a desk. Each place size in the box was 12-by-10 cm. The assignment of each of the eight items to a place was counterbalanced across participants. (2) Test: all children pseudo randomly received eight study items (banana, spoon, dog, cap, car, block, button, and scissors) and three non-studied items as distractors (cup, toothbrush, and airplane).

Procedure

The procedure consisted of three phases (see Figure 2). In the first phase, the experimenter taught the eight study items by showing them. After that, the experimenter instructed the children to carefully watch where each small toy was placed. All children were instructed what the answer was by being shown each toy in its place. At this time, each toy was removed from its place immediately after the experimenter put it there. As an immediate check at that time, the experimenter checked whether the children could overtly show the place of each toy by pointing to the correct area. If they pointed to the incorrect area, the experimenter taught the correct area by actually putting each toy in its place. Moreover, the experimenter gave the correct feedback again even when children showed the correct area. This manipulation was conducted in order to prevent errors in learning.

In the second phase, all the children received the 4 repeated retrieval (4RRR) toys and the 4 repeated study (4SSS) toys alternately one by one. These were counterbalanced so that half of the participants engaged in placing the repeated

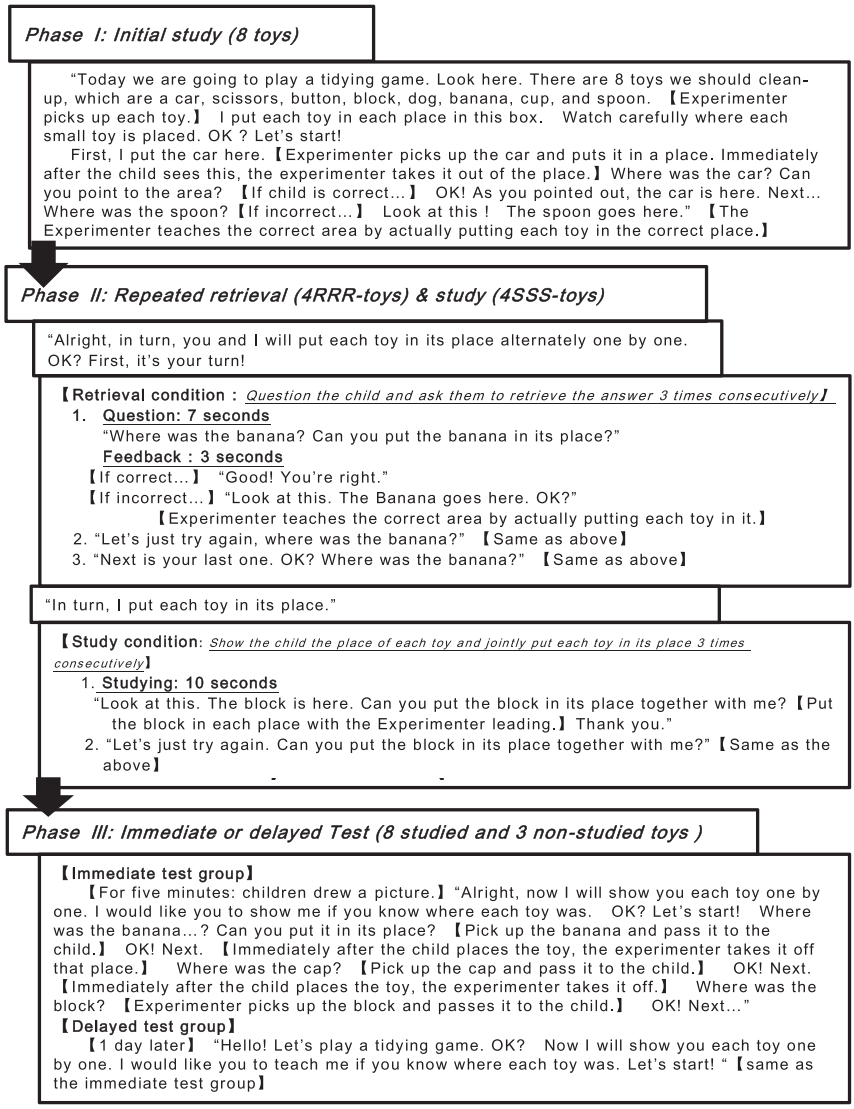


Figure 2. The flow of experiment.

retrieval toys first, and the other half engaged in placing the repeated study toys first. They were asked to report overtly by showing the place of each toy. Each toy was removed by the experimenter from its place immediately after a child put it there. In the repeated retrieval condition, the experimenter asked them to put each toy one by one in its place three consecutive times. After each time the experimenter gave feedback at once by actually putting each toy in its correct place and asking the child to put it in the correct area again, if they did not know the answer or put the toy in the wrong place. In the repeated study condition,

children were asked to put each toy in its place jointly with the experimenter leading them, three consecutive times. Children were given 7-s to put each toy in its place and 3-s for feedback in the repeated retrieval condition and 10-s to put each toy in its place in the repeated study condition. Thus, we provided a regular timed schedule in both the retrieval and study conditions. The eight toys were assigned randomly to each condition.

In the third phase, children in the immediate test group were asked to put each toy in its place after waiting for 5 min. Similarly, children in the delayed test group were asked to put each toy in its place after one day. All the children were able to place each toy at their own pace. The assignment of questions for each test was randomly rearranged from the initial study session.

Results

Although recall performance on the immediate check during initial study (phase I) was high (6-year-olds: repeated restudy items (.98) and repeated retrieval items (.99); 5-year-olds: repeated restudy items (.99) and repeated retrieval items (.99), a total of four children (two children each in the repeated study and retrieval conditions) who failed the immediate check by not knowing the location of items even after just being shown were excluded from the analysis below.

First, we analyzed the number of correctly rejected non-studied distractor items. The analysis showed that the mean correct rejection performance in the immediate testing group ($M = .78$, SD = .40) was better than that in the delayed testing group ($M = .40$, SD = .41; $t\,(54) = 3.57$, $p < .001$, $r = .43$). Thus, the accuracy for correctly rejecting non-studied items after the delay was decreased by half, compared to immediate testing.

Figure 3 shows mean positional recall performance as a function of Learning type, and Retention interval.

A 2 (Learning condition: Retrieval vs. Study) × 2 (Retention interval: Immediate vs. Delayed) mixed factorial ANOVA was conducted. The main effect of Learning condition was significant ($F\,(1, 54) = 23.21$, MSE = .24, $p < .001$, $\eta^2 = .04$). Mean positional recall performance in the repeated retrieval condition (.77) was better than that in the repeated study condition (.66). Moreover, the main effect of Retention interval was significant ($F\,(1, 54) = 85.76$, MSE = .24, $p < .001$, $\eta^2 = .52$). The positional recall performance in the 1 day later group (.51) was worse than that in the immediate group (.92). More importantly, the interaction effect of Learning condition × Retention interval was significant ($F\,(1, 54) = 16.38$, MSE = .24, $p < .001$, $\eta^2 = .03$). In the immediate test group, the positional recall performance of the repeated retrieval condition did not differ from that in the repeated study condition ($F\,(1, 54) < 1$, ns, $\eta^2 = .00$), whereas in the 1 day later group, the recall performance of the repeated retrieval condition was better than that in the repeated study condition ($F\,(1, 54) = 39.29$, MSE = .24, $p < .001$,

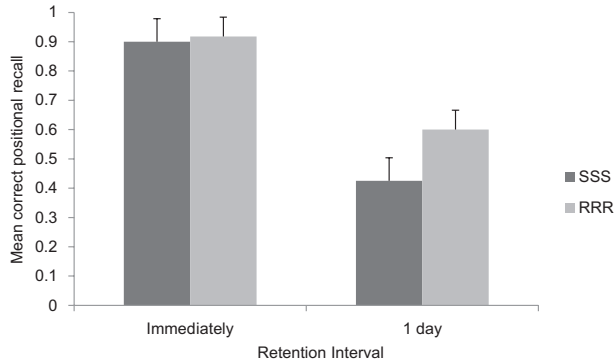


Figure 3. The mean positional recall performance of studied items as a function of Learning type and Retention interval.
Notes: SSS = repeated studying condition, RRR = repeated retrieval condition.

$\eta^2 = .07$). The results showed repeated retrieval benefited learning retention, in spite of the fact that accuracy decreased after delay.

Discussion

The results indicated that retrieval during studying was more beneficial than repeated study for the long-term retention of nonverbal materials for preschool children, irrespective of the finding that the accuracy after delay decreased. As shown in Figure 3, positional performance in the immediate testing group for the retrieval condition did not differ from that for the re-study condition, but results from the delayed test show the advantage of retrieval over studying. A similar pattern of results for short retention intervals has been reported in other studies (e.g., Coppens, Verkoeijen, & Rikers, 2011; Kornell, Bjork, & Garcia, 2011; Roediger & Karpicke, 2006a, 2006b; Toppino & Cohen, 2009; Wheeler, Ewers, & Buonanno, 2003). The observed findings in this study that the testing effect increases with longer retention intervals is also consistent with other studies (e.g., Roediger & Karpicke, 2006a, 2006b).

The current study suggests that repeated retrieval can be an effective strategy for preschool children. These results extend previous findings where verbal materials were used with adults (e.g., Roediger & Karpicke, 2006b), elementary school children (e.g., Goossens et al., 2014; Lipowski et al., 2014; Morris et al., 2005), and preschool children (e.g., Fritz et al., 2007). The results of this study also correspond to map learning findings for adults and elementary school children (Carpenter & Kelly, 2012; Carpenter & Pashler, 2007; Kelley et al., 2015). Fritz et al. (2007) showed that repeated retrieval improved the retention of toy names after 1 min and after 1 day for pre-school children over expanded-representation and massed elaboration. Our study using a spatial location task showed

that repeated retrieval was equally beneficial after 5 min, but more beneficial for retention after 1 day than repeated study, thus solidifying and expanding its efficacy for preschool learners and showing that the human brain is primed from a very early age to learn through retrieval effort. To our knowledge, our study is the first demonstration that the powerful effect of repeated retrieval on memory performance 1 day later generalizes to a nonverbal task with preschool children. Thus, it is worth noting that repeated retrieval has a broadly beneficial effect on memory retention for preschool children through to adults in a variety of verbal and spatial tasks.

As stated before, previous studies of repeated retrieval have been mainly restricted to different types of verbal learning. This study was designed to examine whether repeated retrieval improves the retention of location memory for small toys in preschool children. The learning process of such a nonverbal task might be different from that of a verbal task. Firstly, although all learning requires both explicit knowledge and an implicit procedural component to a certain degree, spatial learning in particular relies on the component of implicit memory (for discussion, see Kelley et al., 2015). In the case of most verbal tasks, retrieval overtly plays a role by activating semantic information and forming the mediated information between cue and item (e.g., Carpenter, 2011; Pyc & Rawson, 2010). On the other hand, in spatial tasks, retrieval may implicitly play a role in the memory updating of the changing relation between the learner and the extant spatial environment. The setting in this study included context information about distance and direction of self-motion (e.g., when putting the toy banana in the specific area) and also visual properties (e.g., color and geometry), acceleration, and movement plans.

Secondly, this study focused on the importance of errorless learning in a spatial task (e.g., Maxwell et al., 2001). Previous studies of the testing effect in the verbal area have shown the beneficial effect of feedback about errors after testing (e.g., Butler & Roediger, 2008). However, in the case of spatial learning, feedback after self-testing was found to have a negative effect compared to restudying (Kelley et al., 2015), because spatial learning involves the implicit procedural component, (Gillner & Mallot, 1998; Golledge, 1991; Hartley, Maguire, Spiers, & Burgess, 2003; Kelley et al., 2015) which does not distinguish self-generated errors from correct responses (Page, Wilson, Shiel, Carter, & Norris, 2006). Therefore, during initial learning, we checked whether all the children could put each toy in the correct place. As a result, four children showing response error during the initial studying despite feedback were eliminated from the analysis. Thus, the procedure of an immediate check during initial learning might lead to a more powerful testing effect, although the performance in the immediate test groups appears to be a ceiling effect. Considering that such spatial aspects of episodic memory have been relatively disregarded in both adult's and children's literature, it is of great worth that retrieval is seen as a learning strategy promoting retention in different types of spatial memory.

Moreover, this study included a self-performance task in the procedure during the second phase. In the repeated study condition, all the children put each toy in its place with the experimenter's assistance, whereas in the retrieval condition, children were asked to put each toy in its place without the experimenter's support. As a result, action memory during learning was promoted more by repeated retrieval than by repeated studying. This is consistent with the hypothesis that being tested by having to retrieve a correct response can lead to the exertion of more retrieval effort, compared to that required for studying. Enacting a self-test would require the implicit association of each toy with its context information, such as distance from box edge, visual properties, and self-motion. This idea of spatial learning is consistent with the espousal by Rowland (2014) of the importance of effortful processing to the testing effect, resulting from his meta-analysis of the testing effect, drawing mainly on verbal learning studies. This idea is also consistent with Fritz et al. (2007) in their study of preschool children's ability to remember names of toys; indicating that an expanded retrieval schedule induced retrieval effort and promoted the testing effect benefit more than expanded re-presentation or expanded elaboration.

Our work on location memory has implications for educational settings and methods. Teachers can promote spatial location learning by asking a child to point to where an object should be, providing feedback to correct errors, and then repeatedly encouraging the child to retrieve their knowledge of the appropriate location. Our findings support the view that learning environments where children are repeatedly shown or told information to be learned is not as effective for long-term retention as methods that require students to put their own effort into retrieving information. It can thus be emphasized that testing has a powerful effect on future retention and should be considered to be a learning strategy in its own right. The results of the current spatial location study highlight the utility of repeated retrieval as an effective learning strategy for retaining information needed in daily life even for preschool children.

From a developmental psychology perspective, spatial location skill development is closely related to broader areas of development (e.g., Newcombe, 2010; Spelke et al., 2010; Uttal et al., 2013). It is thus important to develop methods that can improve spatial memory throughout life. Further research is necessary to examining the optimal conditions for retrieval and retention and the transfer of spatial learning for different ages. These investigations can shed light on not only the practical implications for making use of the benefits of repeated retrieval, but also the mechanisms behind testing effects in regard to potential developmental changes.

Disclosure statement

No potential conflict of interest was reported by the authors.

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